

Difference Between Structured and Unstructured Data

Feature	Structured Data	Unstructured Data
Definition	Data organized in a fixed format (rows & columns)	Data without predefined format or structure
Storage	Stored in databases or tables	Stored in files, media, documents
Schema	Predefined schema	No predefined schema
Easy to Analyze?	✅ Easy to query and analyze	❌ Difficult to analyze directly
Tools Used	SQL, Excel, RDBMS	NLP, AI, Big Data tools
Examples	Customer records, sales data, transaction data	Emails, images, videos, social media posts, audio files

Difference between Data Lake vs Data Warehouse vs Data Mart?

Feature	Data Lake	Data Warehouse	Data Mart
Data Type	Raw data	Processed data	Department data
Structure	Any format	Structured	Structured
Schema	Schema-on-Read	Schema-on-Write	Predefined
Users	Data Scientists	Analysts/BI Users	Business Teams
Size	Very Large	Large	Small
Purpose	Storage	Analytics	Department Reporting

Difference between DATESBETWEEN and DATESINPERIOD functions?

Feature	DATESBETWEEN	DATESINPERIOD
Purpose	Returns dates between two specific dates	Returns dates for a relative time period
Date Selection	Manual start & end date	Dynamic period based on interval

Syntax	DATESBETWEEN(DateColumn, StartDate, EndDate)	DATESINPERIOD(DateColumn, StartDate, Number, Interval)
Working Style	Fixed date range	Time intelligence calculation
Date Control	User defines exact dates	Based on months, years, quarters, days
Dynamic Analysis	Less dynamic	Highly dynamic
Common Usage	Custom date filtering	Rolling totals, YoY, Last N months
Example Scenario	Sales from Jan 1 to Mar 31	Last 3 months sales
Best Use Case	Specific date reporting	Time-based trend analysis

Difference between calculated columns and measures?

Feature	Calculated Column	Measure
Definition	New column created using DAX	Calculation performed dynamically
Calculation Time	Calculated during data load/refresh	Calculated at query time
Storage	Stored physically in model	Not stored (calculated on demand)
Memory Usage	Consumes memory	Does not consume extra memory
Context Used	Row Context	Filter Context
Result Type	Value for each row	Single aggregated value
Usage	Used in slicers, filters, relationships	Used in visuals & aggregations
Performance	Slower for large data models	Faster & optimized
Example Use	Profit per row	Total Profit, Average Sales
Best For	Row-level calculations	Aggregations & KPIs

Difference between Append and Merge?

Feature	Append Query	Merge Query
Purpose	Combines tables vertically	Combines tables horizontally
Operation	Adds rows	Adds columns
SQL Equivalent	UNION	JOIN
Structure Requirement	Same columns required	Common key column required
Result	One table with more rows	One table with additional columns
Relationship Needed	Not required	Required (Key column)
Use Case	Combine monthly/yearly files	Lookup or match data
Example Scenario	Jan Sales + Feb Sales	Orders + Customer Details

Difference between dashboard and report?

Feature	Dashboard	Report
Definition	Single-page visual summary	Multi-page detailed analysis
Pages	One page	Multiple pages
Purpose	Quick monitoring	Detailed data exploration
Interactivity	Limited	Highly interactive
Data Source	Multiple datasets	Usually single dataset
Audience	Managers / Executives	Analysts / Business users
Usage	KPI tracking	Data analysis
Example	CEO KPI view	Sales performance analysis

Difference between SUM and SUMX?

Feature	SUM	SUMX
Type	Aggregation Function	Iterator Function
Working	Adds values directly	Calculates row by row then sums
Row Context	Not used	Uses Row Context
Syntax	SUM(Column)	SUMX(Table, Expression)
Performance	Faster	Slightly slower
Use Case	Simple totals	Calculated totals

Difference between Duplicate & Reference?

Feature	Duplicate	Reference
Definition	Creates full copy of query	Creates linked query
Dependency	Independent	Depends on original query
Data Changes	Does not affect original	Updates automatically
Memory Usage	Higher	Lower
Best Use	Separate transformations	Reusing cleaned data
Relationship	No connection	Connected to source

Difference Slicer & Filter?

Feature	Slicer	Filter
Definition	Visual tool used to filter data	Backend filtering option
Visibility	Visible on report canvas	Usually hidden in filter pane
User Interaction	Users can click/select values	Applied by developer or user
Type	Visual object	Filtering functionality

Usage	Interactive filtering	Control data scope
Best For	Easy user selection	Advanced filtering
Example	Year dropdown	Page/Report level filter

Difference between Drilldown and DrillThrough in Power BI?

Feature	Drill Down	Drill Through
Purpose	Navigate within same visual hierarchy	Navigate to another report page
Navigation	Same chart	Different page
Data Level	Hierarchical levels	Detailed record analysis
Example	Year → Quarter → Month	Sales Summary → Customer Detail Page
Setup Required	Hierarchy required	Drill-through filter required
User Action	Click drill icon	Right-click data point

Difference between a pie chart and a donut chart?

Feature	Pie Chart	Donut Chart
Shape	Full circular chart	Circular chart with hole in center
Center Area	Filled	Empty (donut hole)
Data Display	Shows part-to-whole relationship	Same as pie but with better spacing
Readability	Harder with many categories	Easier to read labels
Extra Information	Cannot display center value	Can show KPI/Total in center
Visual Appearance	Simple traditional look	Modern dashboard look
Best Use	Few categories (3–5)	Dashboard visualization

Difference between Power BI Desktop and Power BI Service?

Feature	Power BI Desktop	Power BI Service
Definition	Local application used to create reports	Cloud-based online platform
Installation	Installed on computer	Accessed via web browser
Usage	Report development & modeling	Sharing & collaboration
Data Modeling	Full modeling capability	Limited modeling
Report Creation	Yes	Limited editing
Publishing Reports	Publish to Service	Already hosted reports
Collaboration	Not available	Available
Data Refresh	Manual refresh	Scheduled refresh
Sharing Reports	Cannot share directly	Share with users/workspaces
Best For	Developers / Analysts	Business users / Teams
Example Access	Desktop software	Power BI Service

Difference between referencing and duplicating a query in Power Query.

Feature	Duplicate Query	Reference Query
Definition	Creates a full copy of the query	Creates a new query linked to original
Connection to Original	Independent	Dependent on original query
Data Updates	Changes do NOT affect original	Updates automatically from original
Transformation Sharing	Not shared	Inherits transformations

Memory Usage	Higher	Lower
Performance	Slightly heavier	More efficient
Best Use Case	Need separate dataset	Reuse cleaned/base data
Example	Two independent tables	One base query → multiple outputs

Difference between Tile vs Report vs Dashboard ?

Feature	Tile	Report	Dashboard
Definition	Single visual snapshot	Detailed data analysis pages	Collection of tiles on one page
Level	Smallest element	Detailed level	High-level overview
Pages	No pages	Multiple pages	Single page
Content	One visual (chart/KPI/card)	Multiple visuals	Multiple tiles
Data Source	From report or dataset	Usually one dataset	Multiple reports/datasets
Interactivity	Limited	Highly interactive	Limited interaction
Purpose	Highlight KPI	Deep analysis	Quick business monitoring
User Type	Executives/Users	Analysts	Managers/Leadership
Location	Exists inside dashboard	Created in Desktop	Created in Service

What is ALL and ALLEXCEPT function?

Feature	ALL()	ALLEXCEPT()
Purpose	Removes all filters	Removes all filters except specified columns

Filter Context	Clears entire filter context	Keeps selected column filters
Usage	Calculate overall totals	Calculate totals while preserving some filters
Syntax	<code>ALL(Table/Column)</code>	<code>ALLEXCEPT(Table, Column1, Column2...)</code>
Flexibility	Less selective	More selective
Common Use	Grand Total calculation	Category-wise comparison
Example	Ignore slicers completely	Keep Region filter but ignore others

What do you mean by MTD , QTD & YTD?

Feature	MTD (Month-To-Date)	QTD (Quarter-To-Date)	YTD (Year-To-Date)
Definition	Total from start of current month to today	Total from start of current quarter to today	Total from start of year to today
Time Period	Monthly	Quarterly	Yearly
Calculation Range	1st day of month → Current date	Quarter start → Current date	Jan 1 → Current date
Purpose	Monthly performance tracking	Quarterly analysis	Annual performance tracking
Business Use	Monthly sales	Quarterly targets	Yearly KPIs
DAX Function	<code>TOTALMTD()</code>	<code>TOTALQTD()</code>	<code>TOTALYTD()</code>

What is Difference between MyWorkspace and Workspace?

Feature	My Workspace	Workspace
Definition	Personal workspace for individual use	Shared workspace for team collaboration
Purpose	Personal development & testing	Team projects & production reports

Access	Only owner (private)	Multiple users can access
Collaboration	Not supported	Supported
Sharing Reports	Limited sharing	Easy sharing & collaboration
User Roles	No roles	Admin, Member, Contributor, Viewer
App Publishing	Not available	Available
Best Use	Practice & personal reports	Organizational reporting
Typical Users	Individual developer	Team/Organization

What is the difference between the MAX and MAXX aggregate functions?

Feature	MAX	MAXX
Type	Aggregate Function	Iterator Function
Purpose	Returns maximum value from a column	Evaluates expression row by row and returns maximum result
Row Context	Not used	Uses Row Context
Syntax	MAX(Column)	MAXX(Table, Expression)
Calculation	Direct column aggregation	Calculation per row first
Performance	Faster	Slightly slower
Use Case	Find highest value in column	Find highest calculated value
Example	Max Salary	Max Profit (Sales – Cost)

Difference Between ROW_NUMBER, RANK, and DENSE_RANK

Feature	ROW_NUMBER()	RANK()	DENSE_RANK()
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Purpose	Assigns unique number to each row	Gives same rank for ties	Gives same rank without gaps
Duplicate Values	Not considered	Same rank for duplicates	Same rank for duplicates
Rank Skipping	No skipping	Skips ranks	No skipping
Sequence	Always continuous	May have gaps	Continuous ranking
Use Case	Unique row numbering	Competition ranking	Category ranking
Example	Pagination	Leaderboard	Dense ranking

What is a Star Schema vs Snowflake Schema?

Feature	Star Schema	Snowflake Schema
Structure	Central fact table connected to dimension tables	Fact table connected to normalized dimension tables
Shape	Star-like structure	Snowflake-like structure
Normalization	Denormalized dimensions	Normalized dimensions
Number of Tables	Fewer tables	More tables
Query Performance	Faster queries	Slightly slower queries
Complexity	Simple design	Complex design
Storage Usage	More storage	Less storage
Joins Required	Fewer joins	More joins
Best For	Reporting & BI tools	Large enterprise systems

What's the difference between Fact and Dimension Tables?

Feature	Fact Table	Dimension Table
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Definition	Stores measurable business data	Stores descriptive information
Data Type	Numeric values	Text/attributes
Purpose	Analysis & calculations	Context for analysis
Contains	Metrics/KPIs	Descriptions
Size	Large	Smaller
Examples	Sales Amount, Quantity	Customer, Product, Date
Keys	Foreign Keys	Primary Keys
Aggregation	Used for SUM, COUNT	Used for filtering/grouping

Difference Between YoY and YTD

Feature	YoY (Year-over-Year)	YTD (Year-to-Date)
Definition	Compares performance with the same period last year	Calculates total from start of year to current date
Purpose	Measure growth or decline	Track cumulative yearly performance
Comparison Type	Year vs Previous Year	Current year progress
Time Calculation	Current period vs last year same period	Jan 1 → Today
Business Use	Growth analysis	Performance tracking
Example	Jan 2025 vs Jan 2024 sales	Total sales from Jan → Today
DAX Function	<code>SAMEPERIODLASTYEAR()</code>	<code>TOTALYTD()</code>

Row Context vs Filter Context

Feature	Row Context	Filter Context
Definition	Calculates data row by row	Calculates data based on applied filters

Where it exists	Calculated Columns, Iterators (SUMX, AVERAGEX)	Measures, Visuals, Slicers, CALCULATE
Affects	Current row only	Entire dataset based on filter
Created by	Automatically in calculated column	Report filters, slicers, visuals, CALCULATE
Example	<code>Sales[Qty] * Sales[Price]</code>	<code>SUM(Sales[Amount])</code> filtered by Year
Performance impact	Can increase model size (stored)	Calculated dynamically

What is the difference between Power Query and Power Pivot?

Feature	Power Query	Power Pivot
Purpose	Data extraction & transformation (ETL)	Data modeling & analysis
Main Function	Clean and prepare data	Create relationships & calculations
Stage Used	Before loading data	After loading data
Language Used	M Language	DAX (Data Analysis Expressions)
Data Operations	Filter, merge, append, clean	Measures, calculated columns
Relationships	Not mainly used	Create table relationships
Output	Loads transformed data	Analytical data model
Usage	Data preparation	Data analysis
Example Tool	Transform Data Editor	Model View

Import vs DirectQuery vs Live Connection vs Dual Mode

Feature	Import Mode	DirectQuery Mode	Live Connection	Dual Mode
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Data Storage	Data stored inside PBIX	Data stays in source	Data stays in external model (SSAS / Power BI Dataset)	Can act as Import or DirectQuery
Performance	 Fastest	Depends on source DB	Fast (depends on external model)	Optimized hybrid
Data Size Limit	1 GB (Pro), higher in Premium	No dataset size limit in PBIX	Depends on external model	Depends on setup
Refresh Required	Yes	No (real-time queries)	No (managed externally)	Yes (for import part)
DAX Flexibility	Full DAX support	Some limitations	Limited modeling changes	Flexible
Best For	Medium datasets	Large real-time datasets	Enterprise centralized models	Composite models

RLS, OLS, Column Level Security (CLS), and Page Level Security (PLS)

Feature	RLS (Row-Level Security)	OLS (Object-Level Security)	CLS (Column-Level Security)	PLS (Page-Level Security)
Full Form	Row-Level Security	Object-Level Security	Column-Level Security	Page-Level Security
Controls What?	Rows of data	Entire table or object	Specific columns	Report pages
Level Applied On	Data level	Model object level	Column level	Report level
Where Configured?	Power BI Desktop (Manage Roles)	Tabular Editor	Tabular Editor	Power BI Desktop
Example	User sees only their region data	User cannot see a specific table	User cannot see salary column	User sees only assigned report pages
Visible in Model?	Yes (structure visible, data filtered)	No (hidden object not visible)	Column hidden	Page hidden

Use Case	Region-wise access	Hide sensitive tables	Hide sensitive columns (Salary, Cost)	Department-wise page access
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What are the different types of relationships in Power BI?

Power BI supports **three types of relationships**:

Type	Description	Example
One-to-One (1:1)	One record in Table A relates to one record in Table B	One employee → One ID card
One-to-Many (1:N)	One record in Table A relates to many records in Table B	One customer → Many orders
Many-to-Many (M:N)	Records in both tables relate to multiple records	Many students → Many courses

CALCULATE vs CALCULATETABLE vs FILTER

Feature	CALCULATE	CALCULATETABLE	FILTER
Purpose	Modifies filter context for a scalar value	Modifies filter context for a table	Returns filtered rows from a table
Return Type	Single value (Number/Text)	Table	Table
Usage	Measures & aggregations	Table calculations	Apply row-level filtering
Context Change	Yes	Yes	No context change directly
Common Use	KPI calculations	Virtual tables	Custom filtering logic
Syntax	<code>CALCULATE(Expression, Filter)</code>	<code>CALCULATETABLE(Table, Filter)</code>	<code>FILTER(Table, Condition)</code>
Performance	Optimized	Optimized	Slower if misused

Difference Between ALL, ALLEXCEPT, and ALLSELECTED

Feature	ALL()	ALLEXCEPT()	ALLSELECTED()
Purpose	Removes all filters	Removes filters except specified columns	Keeps user selections but ignores visual filters
Filter Removal	Entire table/column	All except chosen columns	Removes internal visual filters only
User Slicer Impact	Ignored	Partially respected	Respected
Context Behavior	Grand total calculation	Category-wise totals	Visual total based on selection
Return Type	Table	Table	Table
Common Use	% of Grand Total	% within category	% of Selected Total
Best Scenario	Ignore all filters	Preserve one dimension	Respect slicer selection

Difference Between VALUES() and DISTINCT()

Feature	VALUES()	DISTINCT()
Purpose	Returns unique values from a column/table	Returns unique values from a column/table
Blank Handling	Includes Blank row (if exists)	Does NOT add extra blank row
Relationship Impact	Considers model relationships	Ignores relationship blank row
Usage	Dynamic calculations & measures	Data deduplication
Return Type	Table	Table
Common Use	Slicers, measures, context analysis	Remove duplicate values
Behavior with Missing Keys	Adds blank for unmatched data	No additional blank

Difference Between EARLIER() and EARLIEST()

Feature	EARLIER()	EARLIEST()
Purpose	Returns value from previous row context	Returns value from first (outermost) row context
Row Context	Accesses outer row context	Accesses earliest available row context
Usage	Nested row calculations	Multiple nested row contexts
Common Scenario	Ranking, comparisons	Deep nested calculations
Complexity	More commonly used	Rarely used
Where Used	Calculated Columns	Complex Calculated Columns
Syntax	EARLIER(Column)	EARLIEST(Column)

Difference Between RELATED() and RELATEDTABLE()

Feature	RELATED()	RELATEDTABLE()
Purpose	Fetches a single value from related table	Returns related rows as a table
Relationship Direction	Many → One	One → Many
Return Type	Single value	Table
Usage	Calculated Column	Measures / Table expressions
Common Use	Get lookup value	Aggregate related records
Example	Get Customer Name in Sales table	Get all Sales for Customer

Difference Between TOTALYTD(), TOTALQTD(), TOTALMTD()

Feature	TOTALMTD()	TOTALQTD()	TOTALYTD()
Meaning	Month-To-Date	Quarter-To-Date	Year-To-Date
Calculation Range	Start of month → Today	Quarter start → Today	Year start → Today

Business Use	Monthly tracking	Quarterly analysis	Annual performance
Time Period	Monthly	Quarterly	Yearly
Common KPI	Monthly Sales	Quarterly Revenue	Annual Revenue

Difference Between IF() and SWITCH() (DAX)

Feature	IF()	SWITCH()
Purpose	Evaluates one logical condition	Evaluates multiple conditions
Best Use	Simple conditions	Multiple condition checks
Readability	Complex with nested IF	Cleaner and readable
Syntax	<code>IF(condition, true, false)</code>	<code>SWITCH(expression, value, result)</code>
Performance	Slower if nested many times	Better for multiple cases
Example Use	Pass/Fail	Grade classification

Difference Between BLANK() and ISBLANK()

Feature	BLANK()	ISBLANK()
Purpose	Returns an empty value	Checks if value is empty
Type	Value function	Logical function
Return	Blank value	TRUE/FALSE
Usage	Assign empty result	Validate missing data

Difference Between Report-Level, Page-Level, and Visual-Level Filters

Filter Type	Scope	Applies To	Visibility	Example Use Case
Report-Level Filter	Entire Report	All pages & visuals	Applies globally	Show data only for Year = 2025 across report

Page-Level Filter	Single Page	All visuals on one page	Page specific	Show Region = West on Sales page
Visual-Level Filter	Single Visual	Only selected visual	Visual specific	Show Top 10 Products in one chart

Difference Between Bookmarks and Drillthrough

Feature	Bookmarks	Drillthrough
Purpose	Saves report view/state	Navigate to detailed report page
Navigation Type	Same page interaction	Move to another page
Used For	Show/Hide visuals, storytelling	Detailed data analysis
Filter Context	Saves filters, slicers, visibility	Passes selected data filters
User Action	Button click	Right-click data point
Interactivity	UI/Presentation control	Data exploration
Typical Use	Toggle views, navigation menus	Summary → Detail analysis
Example	Switch between charts	Sales Summary → Customer Detail page

Difference Between Bar Chart and Line Chart

Feature	Bar Chart	Line Chart
Purpose	Compare values across categories	Show trends over time
Data Type	Categorical data	Continuous/time-series data
Representation	Rectangular bars	Connected data points (line)
Best Use	Category comparison	Trend analysis
Time Analysis	Not ideal	Best for time-based data
Example	Sales by Region	Sales over Months
Visualization Style	Discrete comparison	Continuous movement

Difference Between Live Connection and Extract Connection

Feature	Live Connection	Extract Connection
Data Storage	Data stays in source system	Data stored locally as extract
Performance	Depends on source performance	Faster performance
Data Refresh	Real-time / near real-time	Requires scheduled/manual refresh
Internet Requirement	Continuous connection required	Can work offline after extract
Data Size	No local storage limit	Limited by extract size
Load Time	Slower if source is slow	Faster dashboards
Use Case	Real-time reporting	High-speed analytics
Example Tools	Direct database connection	Imported snapshot data
